



VIT-AP
UNIVERSITY

Module 1: Understanding Cloud Computing

Fundamental Concepts and Models

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RECAP

- **Cloud origins and influences**
 - Introduction
 - Early Concepts and Precursors
 - Development of the Internet
 - Virtualization Technology
 - Grid Computing
 - Birth of Cloud Computing
 - Influences on Cloud Computing
- **Basic Concepts and Terminology, Goals and Benefits**
 - Key Terms and Definitions
 - Cloud Service Models
 - Cloud Deployment Models
 - Cloud Characteristics
 - Business Goals of Cloud Computing
 - Benefits of Cloud Computing
 - Case Studies

OUTLINE

- **Fundamental Concepts and Models**
 - **Roles and boundaries,**
 - **cloud characteristics,**
 - **cloud delivery models,**
 - **cloud deployment models.**
- Summary
- References

Fundamental Concepts and Models

- ❑ Roles and boundaries
 - ❑ Roles

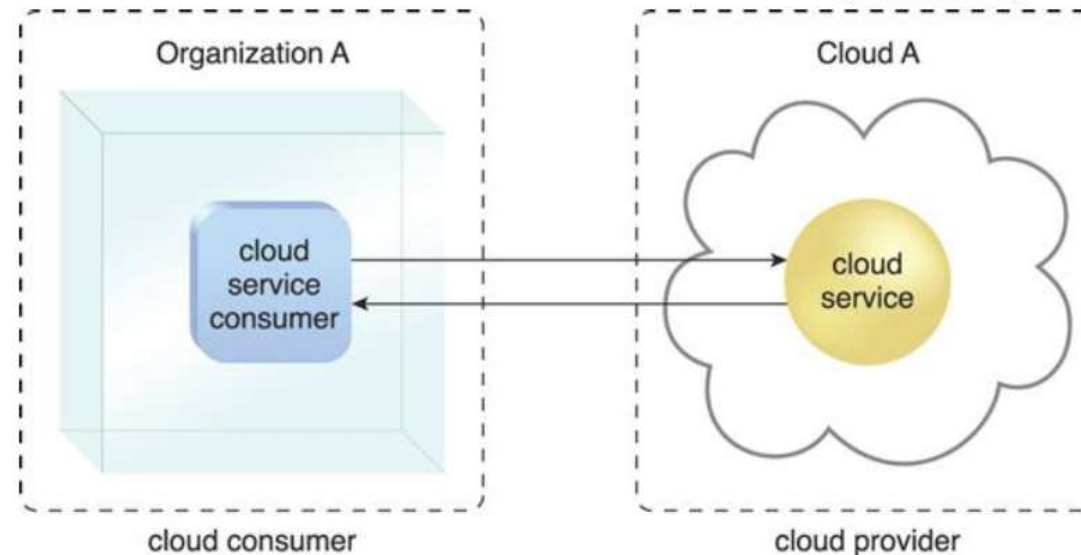
- Roles and boundaries

- Cloud Provider

- Definition: The organization that offers cloud services to consumers.
 - Responsibilities: Manages cloud infrastructure, provides resources and services to consumers.

- Cloud Consumer

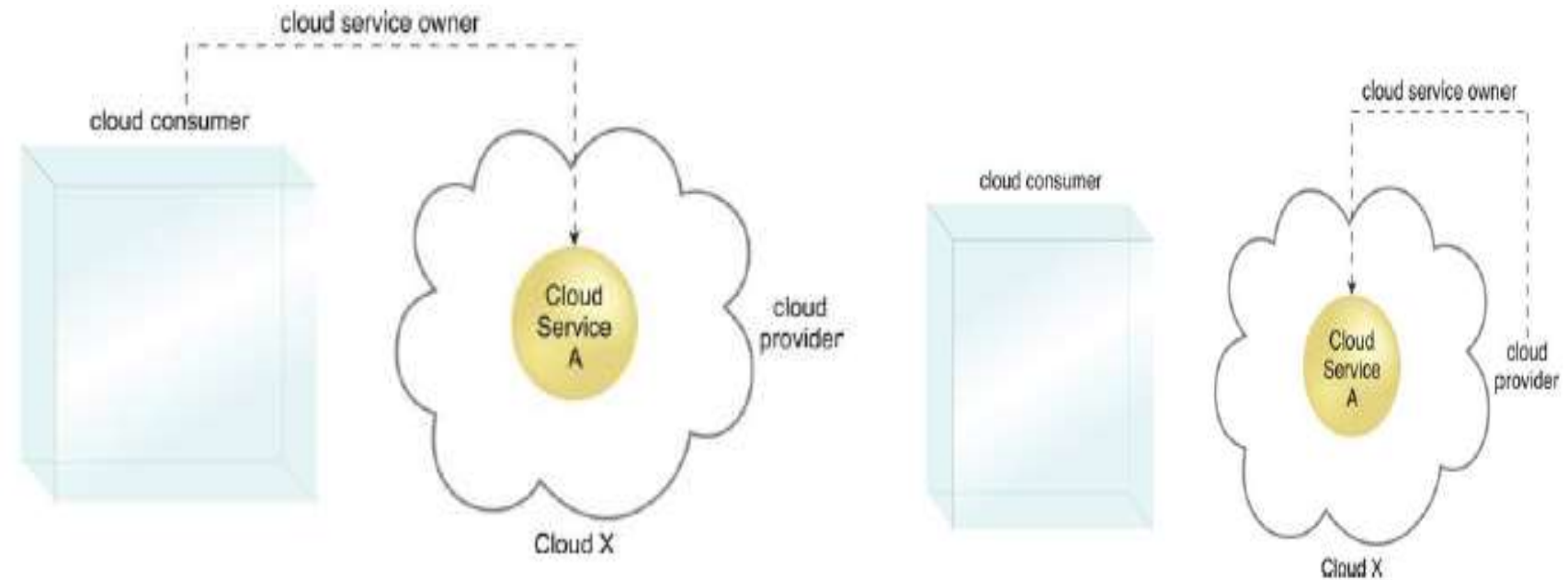
- Definition: Individuals or organizations that use cloud services.
 - Responsibilities: Accesses and utilizes cloud services for various purposes.



Fundamental Concepts and Models

– Cloud Service Owner

- Definition: Entity that legally owns a cloud service.
- Responsibilities: Ensures the service is operational and meets required standards.

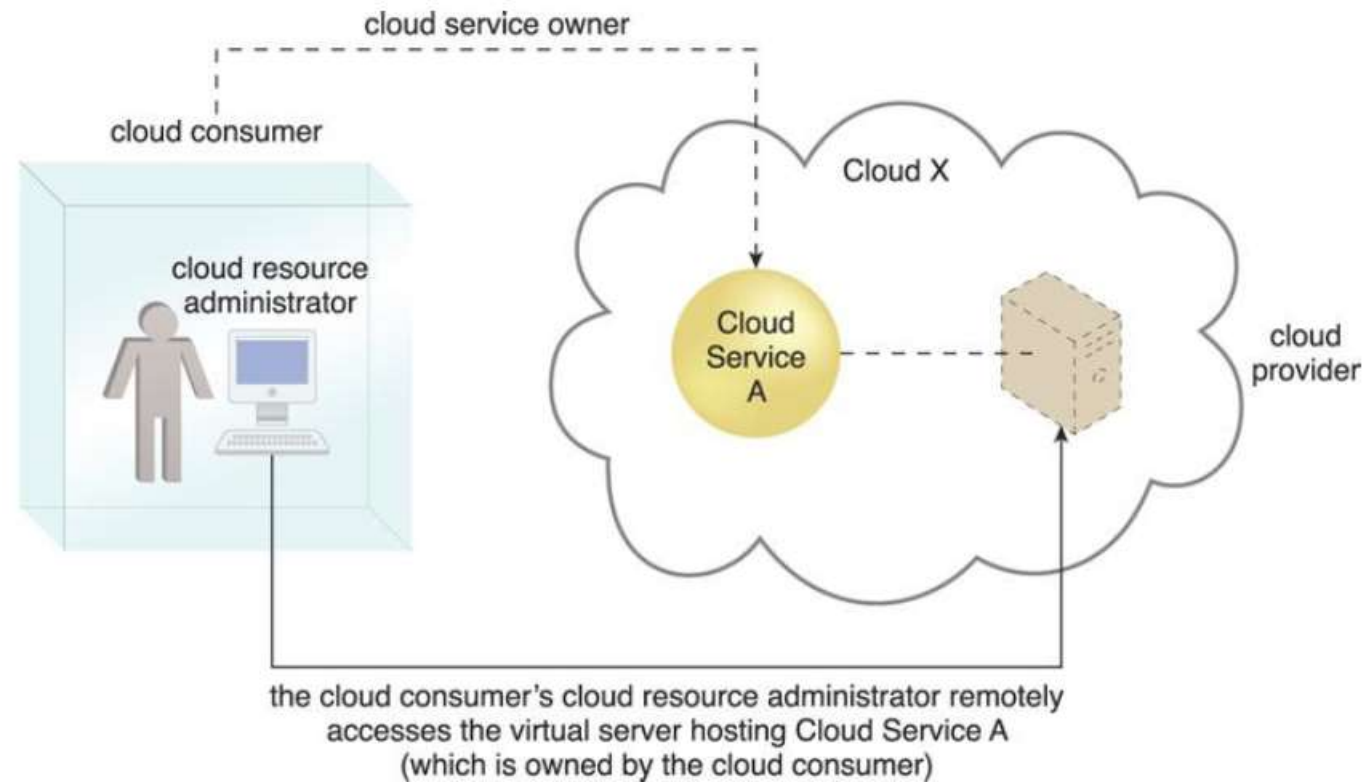


Fundamental Concepts and Models

- Roles and boundaries
- Roles

– Cloud Resource Administrator

- Definition: Manages and maintains cloud resources.
- Responsibilities: Oversees resource allocation, performance, and security.



Fundamental Concepts and Models

- ❑ Roles and boundaries
 - ❑ Roles
 - ❑ Additional Roles

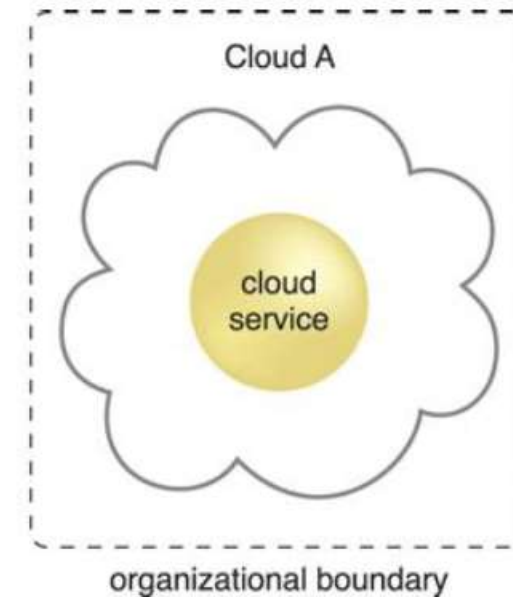
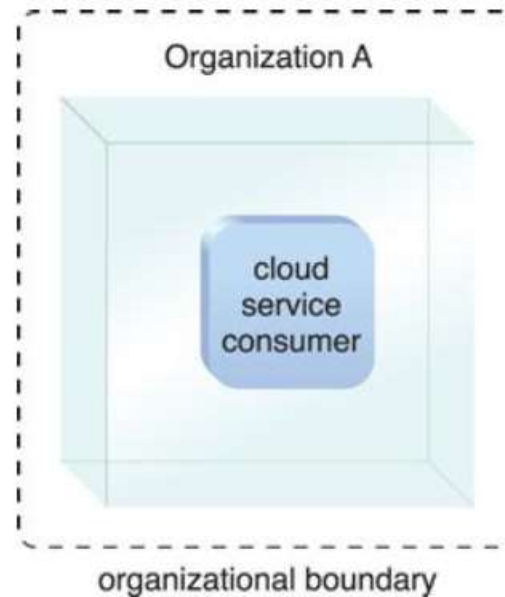
– Additional Roles:

- Cloud Auditor: Conducts independent assessments of cloud services.
- Cloud Broker: Manages the use, performance, and delivery of cloud services.
- Cloud Carrier: Provides connectivity and transport of cloud services.

- ❑ Roles and boundaries
- ❑ Organizational Boundary

Fundamental Concepts and Models

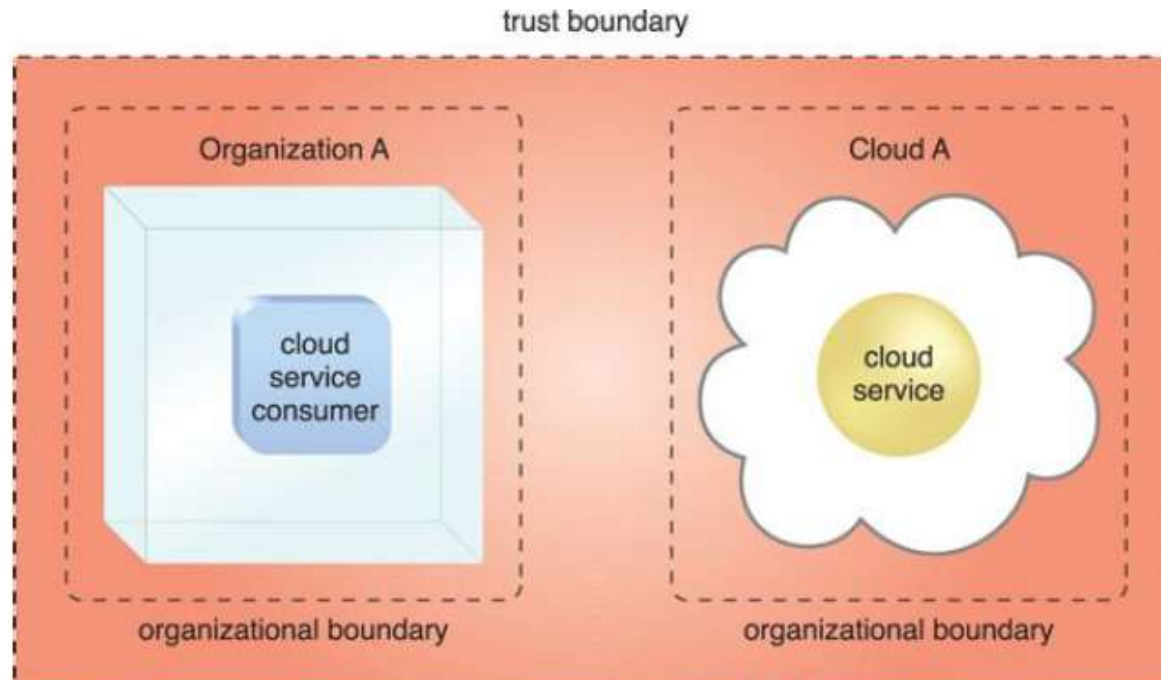
- Organizational Boundary
 - Definition: The physical perimeter that encompasses the IT resources under an organization's control.
 - Example: A company's internal data center.



- ❑ Roles and boundaries
- ❑ Organizational Boundary

Fundamental Concepts and Models

- Trust Boundary
 - Definition: The logical boundary that separates trusted and untrusted resources or users.
 - Example: Division between internal users and external cloud services.



- ❑ Roles and boundaries
- ❑ Organizational Boundary
- ❑ Cloud Characteristics

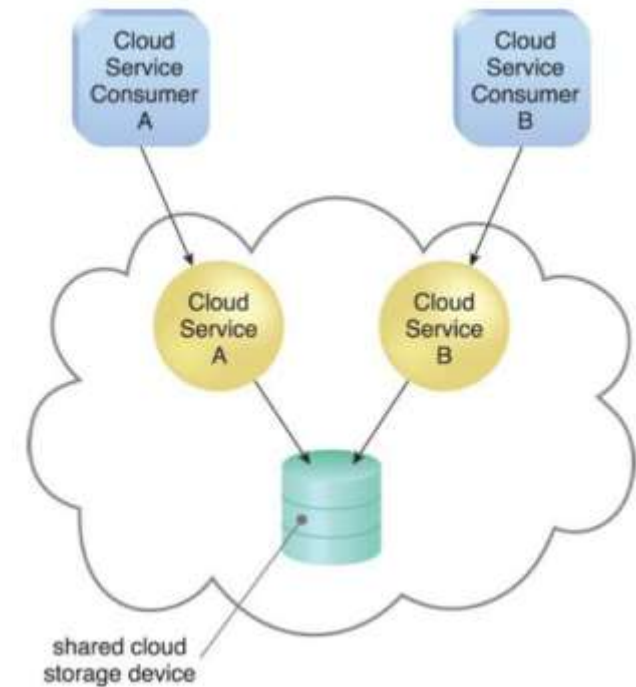
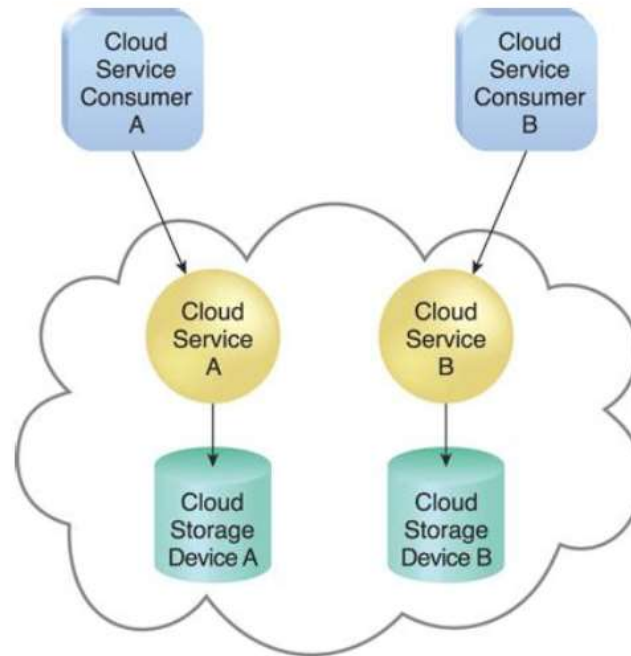
Fundamental Concepts and Models

- Cloud Characteristics
 - On-Demand Usage
 - Self-service provisioning of IT resources.
 - Benefit: Reduces need for human intervention.
 - Ubiquitous Access
 - Wide accessibility of cloud services.
 - Requirements: Support for various devices and protocols.
 - Elasticity
 - Definition: Ability to scale resources automatically.
 - Benefit: Supports varying demand.
 - Measured Usage
 - Definition: Tracking and billing based on resource usage.
 - Benefit: Pay-as-you-go model.

- ❑ Roles and boundaries
- ❑ Organizational Boundary
- ❑ Cloud Characteristics

Fundamental Concepts and Models

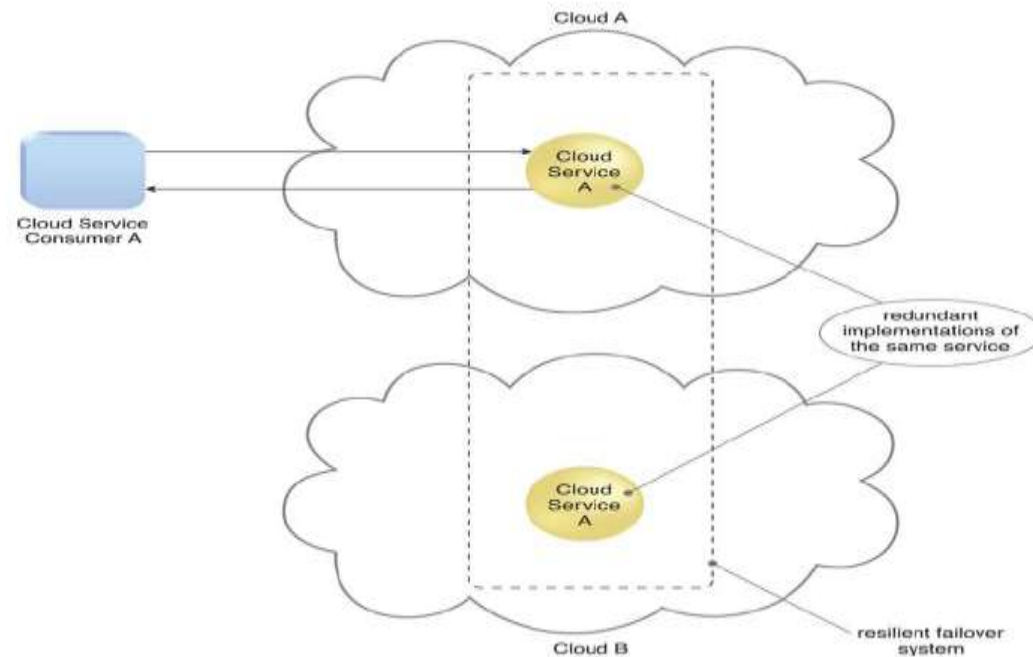
- Cloud Characteristics
 - Multitenancy (and Resource Pooling)
 - Serving multiple consumers with isolated instances.
 - Example: Single-tenant vs. multi-tenant environments.



- ❑ Roles and boundaries
- ❑ Organizational Boundary
- ❑ Cloud Characteristics

Fundamental Concepts and Models

- Cloud Characteristics
 - Resiliency
 - Definition: Failover and redundancy in cloud services.
 - Benefit: Increases reliability and availability.



☐ Roles:

- ☐ Cloud Provider
- ☐ Cloud Consumer
- ☐ Cloud Service Owner
- ☐ Cloud Resource Administrator
- ☐ Additional Roles: Cloud Auditor, Cloud Broker,

☐ Ubiquitous Access

- ☐ Multitenancy (and Resource Pooling)
- ☐ Elasticity
- ☐ Measured Usage
- ☐ Resiliency

☐ Cloud Carrier Boundaries:

- ☐ Organizational Boundary
- ☐ Trust Boundary

☐ Cloud Characteristics

- ☐ On-Demand Usage

References

1. Thomas Erl, Ricardo Puttini, Zaigham Mahmood, “Cloud Computing: Concepts, Technology & Architecture”, PHI Publications, 2013.

Questions



**Thank
You**